MAE 140 Winter 12 Homework #8

T R & T, 11.8, 11.9, 11.10, 11.11, 11.26, 11.28, 11.52, 11.64, 12.14, 12.56

Due: 03/15/2012

FIGURE P11–7

11–8 Find the current transfer function $T_I(s) = I_2(s)/I_1(s)$ in Figure P11–8. Select values of R and L so that $T_I(s)$ has a pole at $s = -500$ rad/s.

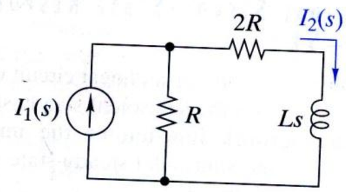

FIGURE P11–8

11–9 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11–9. Locate the poles and zeros of the transfer function.

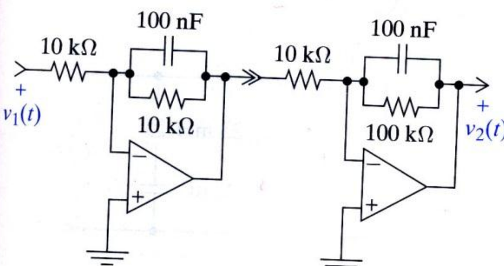

FIGURE P11–9

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11–10 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11–10. Locate the poles and zeros of the transfer function.

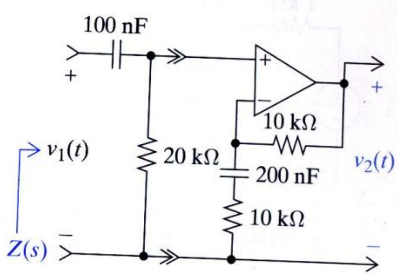


FIGURE P11–10

11–11 Find the input impedance $Z(s)$ in Figure P11–10.

11–26 The circuit in Figure P11–26 is in the steady state with $v_1(t) = 5\cos 225.08t$ V. Find $v_{2SS}(t)$. Repeat for $v_1(t) = 5\cos 10kt$ V. And without doing any calculations, repeat for $v_1(t) = 5$ V.

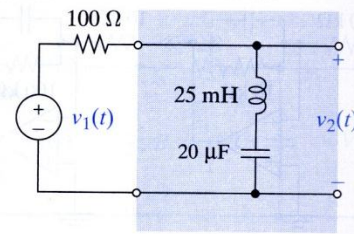


FIGURE P11–26

11–28 The circuit in Figure P11–28 is in the steady state with $i_1(t) = 10\cos 500t$ mA, $R_1 = 100\,\Omega$, $R_2 = 400\,\Omega$, and $L = 100$ mH. Find $i_{2SS}(t)$. Repeat for $i_1(t) = 10\cos 5000t$ mA.

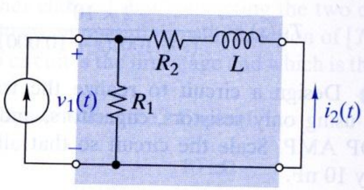


FIGURE P11–28

11–52 Design a circuit to realize the transfer function below using only resistors, capacitors, and no more than one OP AMP. Scale the circuit so that all capacitors are exactly 10 nF.

$$T_V(s) = \pm\frac{100\,(s+1000)}{(s+100)(s+10{,}000)}$$

11–64 A OP AMP Modules and Loading

Figure P11–64 shows an interconnection of three basic OP AMP modules.

(a) Does this interconnection involve loading?

(b) Find the overall transfer function of the interconnection and locate its poles and zeros.

(c) Find the output $v_2(t)$ when the input is $v_1(t) = \cos 500t$ V. Repeat for $v_1(t) = \cos 10\text{k}t$ V and again for $v_1(t) = \cos 200\text{k}t$ V.

(d) Can you think of a use for this circuit?

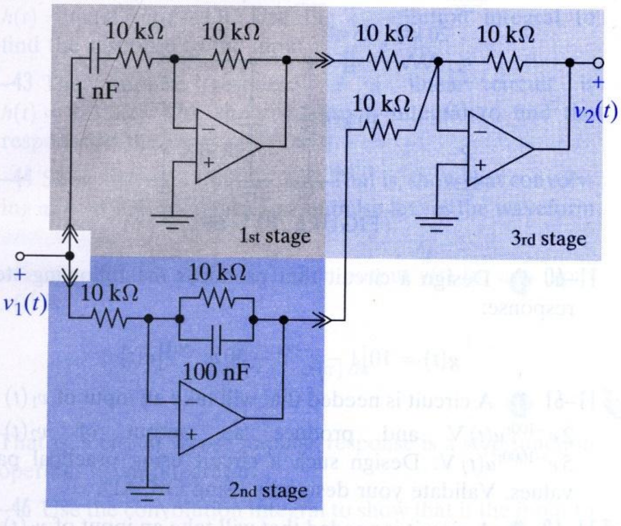


FIGURE P11–64

12–14 D The circuit in Figure P12–14 produces a bandpass response for a suitable choice of element values. Identify the elements that control the two cutoff frequencies. Select the element values so that the passband gain is 30 and the cutoff frequencies are 100 rad/s and 2500 rad/s. Use practical element values with $R \geq 10\,\text{k}\Omega$ and $C \leq 1\,\mu\text{F}$. Use OrCAD to validate your design.

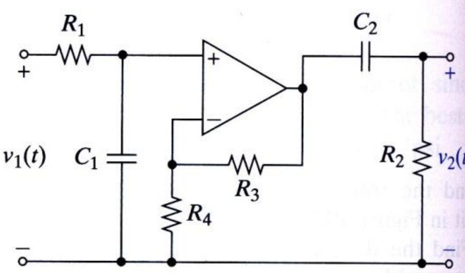


FIGURE P12–14

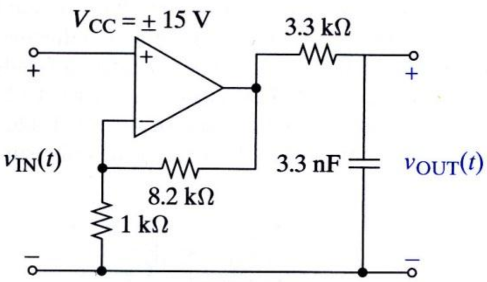
12–56 E Design Evaluation

Your company has issued a request for proposals, listing the following design requirements and evaluation criteria.

Design requirements: Design a low-pass filter with a passband gain of $9 \pm 10\%$ and a cutoff frequency of $90 \pm 10\%$ krad/s. A sensor drives the filter input with a 1-kΩ source resistance and an open-circuit voltage range of ±1.6 V.

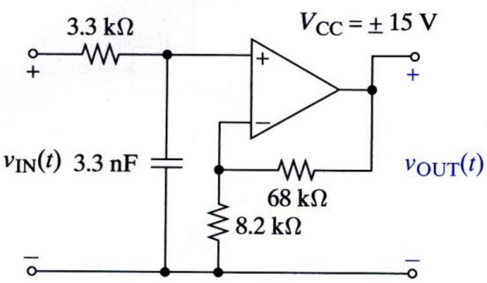
Evaluation criteria: Filter performance, parts count, use of standard parts, and cost. The two vendors have responded with the designs shown in Figure P12–56. As a junior engineer, the project manager asks you evaluate the designs and recommend a vendor. Which vendor would you recommend and why?

First-Order Filter Company



Unit Cost: $29.00

Simply Filters, Ltd



Unit Cost: $20.99

FIGURE P12–56